Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Made with GAMMA

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data columns tracked

37

Missing Values

Review ratings imputed

Customer demographics, purchase details, and shopping behavior captured across product categories

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Python Data Preparation

Data Loading

Imported using pandas

Exploration

info() and describe() for structure and statistics

Missing Data

Median imputation by category

Standardization

Snake case column names

Feature Engineering

Age groups and purchase frequency

Database Load

PostgreSQL integration

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SQL Analysis: Revenue Insights

Gender Revenue

Female: $75,191

Male: $157,890

Age Groups

Young Adult: $62,143

Senior: $55,763

Shipping Types

Standard: $58.46

Express: $60.48

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Product Performance

1

Top Rated

Gloves (3.86), Sandals (3.84), Boots (3.82)

2

Best Sellers

Blouse, Pants, Jewelry (171 orders each)

3

Discount-Heavy

Hat (50%), Sneakers (49.66%), Coat (49.07%)

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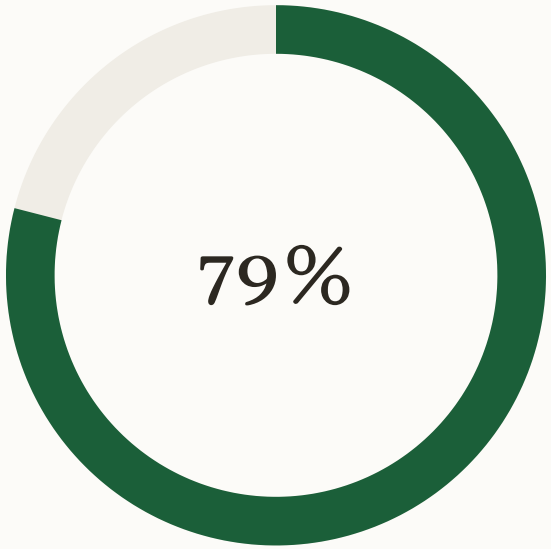
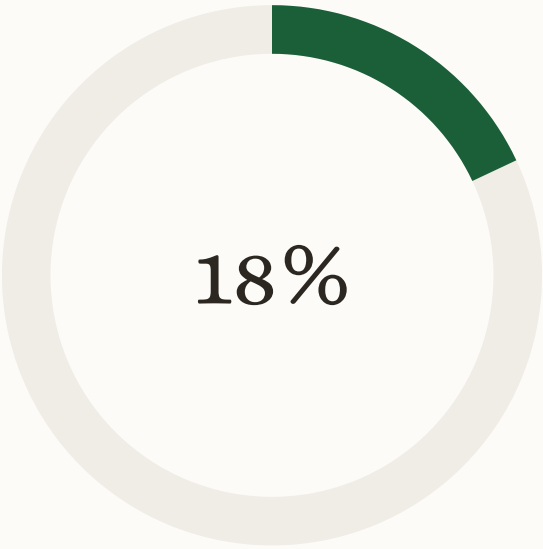
Customer Segmentation



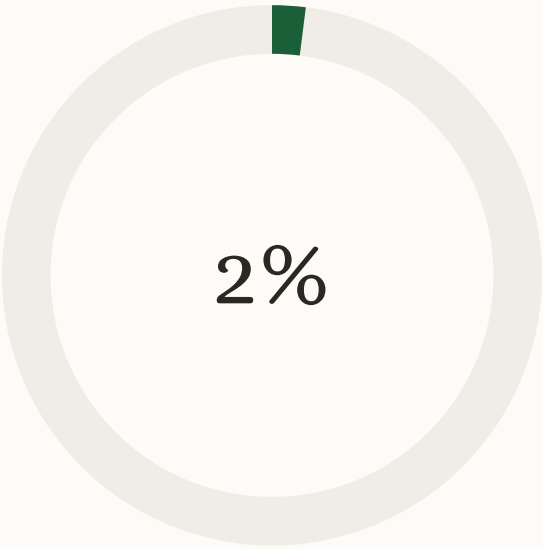
Loyal

3,116 customers



Returning

701 customers



New

83 customers

Classification based on purchase history and frequency patterns

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Subscription Analysis

Subscriber Base

1,053 subscribers (27%) vs. 2,847 non-subscribers (73%)

Revenue Impact

Subscribers: $62,645 | Non-subscribers: $170,436

Average Spend

Subscribers: $59.49 | Non-subscribers: $59.87

Repeat Buyers

958 customers with >5 purchases subscribed

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Power BI Dashboard

01

KPIs Displayed

3.9K customers, $59.76 avg purchase, 3.75 avg rating

02

Interactive Filters

Subscription, gender, category, shipping type

03

Visualizations

Donut charts, bar charts by category and age group

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Key Findings

Male Dominance

2.1x revenue vs. female customers

High Discount Usage

50% of hat purchases discounted

Loyal Base

79% classified as loyal customers

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Strategic Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers

Discount Policy

Balance sales with margin control

Targeted Marketing

Focus on high-revenue segments

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